Errata

On Galos and Coppock (2023), *Science Advances* 9, eade7979

This note lists twelve corrections to "Gender composition predicts gender bias: A meta-reanalysis of hiring discrimination audit experiments," published in *Science Advances* 9, eade7979, doi 10.1126/sciadv.ade7979: three misstated numbers, one dropped significance marker, one cross-reference to a figure that does not exist, two reference-list entries with an author's name broken, one figure that shows imputed categories as though they had been coded, one supplementary figure that prints an estimate clipped short of its own value, two more reference-list entries, one in each of the two places this paper sets its reference list, and one page range that ends a page early in both. It also carries seven entries of a second kind, which quote a passage and propose no change to it: four strings the deposited data supply to the supplement's figures and tables, one misabbreviated journal name this paper sets identically in both printings of its reference list, one sentence whose count of papers the paper's own PRISMA diagram does not bear out, and one year the introduction and the abstract give differently. **No correction and no note here changes a conclusion of the paper.** The gender gradient, the estimates in Tables 1 and 2, and every study-level effect in the supplementary materials are unaffected.

Every number in a corrected sentence below is computed from the deposited replication archive (10.7910/DVN/5MCRNY) at the time this document is rendered, including the numbers that were not wrong. All but one of the six reference-list entries are verified instead against the authoritative record at Crossref, named in each entry. The exception, entry 15, is a number that has lost its period, which no external record speaks to: it is verified against the other 127 entries of the list it sits in and against the article's own printing of the same reference. The code and output supporting each correction are at github.com/active-maintenance-aec/galos_coppock_2023.

Bold marks the corrected token in each corrected sentence. Entries 4 and 6 correct a figure and a table rather than a sentence, so each is named in the entry and shown in a reprint of the float, which is otherwise identical to the published one. Entry 4's corrected sentence is the figure's own caption, which has to change with it: the caption is the figure's only key, and the published one describes three colors where the corrected figure draws four. Percentage-point quantities are given at the precision the article itself uses for them, which in the Results sentence corrected below is two decimal places. Rounding that sentence to one decimal would erase the correction it needs, since the published and corrected figures differ by a hundredth of a point.

1. The record spans six continents, not five

> Since 1983, more than 70 employment audit experiments, carried out in more than 26 countries across **five** continents, have randomized the gender of fictitious applicants to measure the extent of hiring discrimination on the basis of gender.

Corrected:

> Since 1983, more than 70 employment audit experiments, carried out in more than 26 countries across **six** continents, have randomized the gender of fictitious applicants to measure the extent of hiring discrimination on the basis of gender.

The paper's own evidence map, fig. S2.2, is drawn as six continent facets: Africa, Asia, Europe, N. America, Oceania, S. America. Section S2 names Oceania among the regions where the boom in audit experimentation occurred, and the earliest study in the whole record, the Australian one the introduction discusses, is on that continent. The sentence describes the wider literature rather than the 57 studies reanalysed here, but a superset of those studies cannot span fewer continents, so five is too few under any reading.

2. Table S2.1 overstates the occupation counts for two studies

> Gonzalez et al. (89): Spain … **18** occupations
>
> Dahl and Krog (84): Denmark … **24** occupations

Corrected:

> Gonzalez et al. (89): Spain … **17** occupations
>
> Dahl and Krog (84): Denmark … **21** occupations

The article disagrees with itself in these two rows. Figs. S4.37 and S4.32 plot 17 and 21 occupations for the same two studies, and 17 and 21 is what the deposited data hold. The other 55 rows of the column are right. The table is not reprinted here because its N subjects column cannot be regenerated from the deposit, which carries no subject counts.

3. The difference in the magnitude of the two predicted biases

> The magnitude of these biases is higher in women-dominated occupations than in men-dominated occupations, by a statistically significant **2.10** points (bootstrapped SE: 0.58), perhaps suggesting that pro–status quo forces operate differently across contexts.

Corrected:

> The magnitude of these biases is higher in women-dominated occupations than in men-dominated occupations, by a statistically significant **2.11** points (bootstrapped SE: 0.58), perhaps suggesting that pro–status quo forces operate differently across contexts.

The deposited code and the maintained rewrite both give 2.10547, which rounds to 2.11 at the two decimal places the sentence prints. The published figure is the truncation rather than a different number, and the bootstrapped standard error, 0.5781, is 0.58 as published.

The figure

4. Figure 4 colors imputed gender composition categories as though the study authors had given them

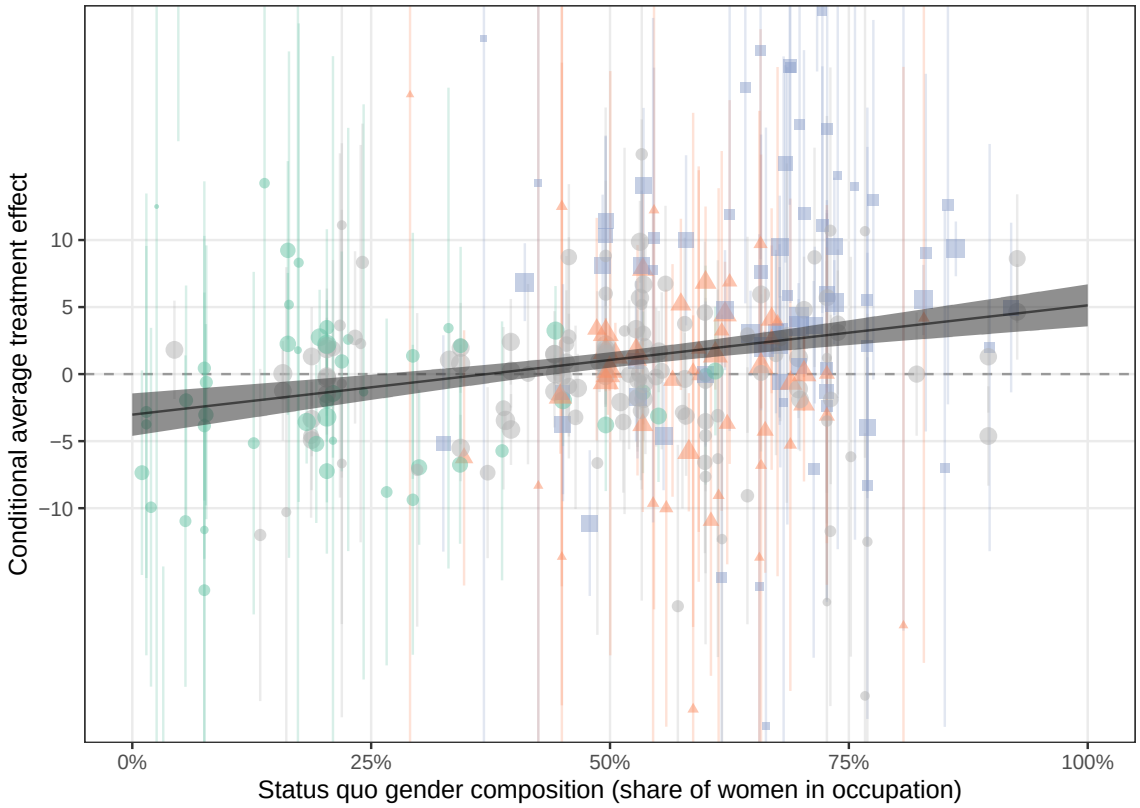
> The gender composition categories (**as given by study authors where available and by us otherwise**) are distinguished by color and shape (green circles, men-dominated; orange triangles, gender-balanced; blue squares, **women-dominated**).

Corrected:

> The gender composition categories (**as given by study authors**) are distinguished by color and shape (green circles, men-dominated; orange triangles, gender-balanced; blue squares, **women-dominated; grey circles, category not given by the study authors**).

Of the 298 study-occupation cells the figure draws, only 174 carry a gender composition category the original study's own authors assigned; the remaining 124 were filled in by us from where the occupation falls in the ILO gender distribution. The caption discloses that in words, so the article is accurate about what it did, but the figure gives all 298 points the same three colors, and a color is a claim about the occupation that a reader cannot tell apart from the 174 that were coded. The corrected figure draws the 124 imputed cells in grey, keeping their position, their interval and their weight. NO ESTIMATE MOVES: the meta-regression is fitted on the continuous share of women in the occupation and never on the category, so Tables 1 and 2, the gender gradient and every study-level effect in the supplementary materials are exactly as published. What changes is which of 298 points is which color. The coding is the same authors' own, carried back from the successor project, where the distinction is kept as a column and the imputed categories are dropped rather than plotted.

Figure 4, corrected. Study-occupation level CATE estimates of signaling applicant is a woman versus a man on callbacks. Point size is proportional to meta-analytic weight. The regression line is derived from the model reported in column 1 of Table 1. The gender composition categories (as given by study authors) are distinguished by color and shape (green circles, men-dominated; orange triangles, gender-balanced; blue squares, women-dominated; grey circles, category not given by the study authors).



Note: The only changes from the published figure are the color of the 124 study-occupation cells whose category the original study authors did not assign, now grey, and the clause naming them in the caption. Every point keeps the position, interval and weight it has in the published figure, and no estimate in the paper moves.

5. The article cites a supplementary figure that does not exist

> The PRISMA search process yielded over 70 papers that met our inclusion criteria. Figure S6.**66** is a "PRISMA flow diagram" that illustrates the search strategy and reports the number of records retained at each stage of the process.

Corrected:

> The PRISMA search process yielded over 70 papers that met our inclusion criteria. Figure S6.**63** is a "PRISMA flow diagram" that illustrates the search strategy and reports the number of records retained at each stage of the process.

The supplementary materials list Figs. S1.1 to S6.63 in their front matter, and the PRISMA flow diagram is the last of them. There is no fig. S6.66. The same paragraph cites the evidence map as fig. S.2.2, where the figure itself is numbered S2.2.

6. Table 2's Model 2 intercept carries no significance marker

The Model 2 intercept is printed as −3.72 with no marker, at a p-value of 0.00104. Five decimals rather than the three the table's own note uses: at three the p-value reads 0.001, which is the threshold for the third star rather than the second, and the correction would be ambiguous inside the rule it is being compared against. The numerically identical Model 3 intercept, −3.73, does carry its two stars. The estimate and its standard error are right and only the marker is missing, so what is corrected here is the table rather than any sentence.

Table 2, corrected. Meta-regression estimates of the gender gradient by majority/minority status. *P < 0.05, **P < 0.01, and ***P < 0.001.

	Model 1	Model 2	Model 3
Intercept	−1.55	−3.72**	−3.73**
	(1.31)	(1.13)	(1.14)
Gender gradient	5.70*	9.08***	9.11***
	(2.50)	(2.26)	(2.28)
Majority status			2.19
			(1.74)
Majority × gradient			−3.42
			(3.37)
Sample	Majority status group	Minority status group	Both groups
Tau.squared	6.60	6.28	6.44
Nobs	91	96	187

Note: The only change from the published table is the marker on the Model 2 intercept. Standard errors in parentheses.

The supplementary figure

The entry below is the only one in this note that no comparison of text layers could have found. A figure's label can overflow its plot panel, and what is then clipped away is ink and not text: the page shows one number and its own text layer carries another, so the published figure and any faithful re-extraction of it agree about a value the reader never sees. It was found by reading the two against each other, with an extractor that applies the clipping path and one that does not.

7. Figure S1.1 prints a clipped estimate for Neumark et al.

Fig. S1.1, Neumark et al. (9): United States: **2.3** (8.4)

Corrected:

Fig. S1.1, Neumark et al. (9): United States: **−12.3** (8.4)

The figure's own text carries the right number and the page does not print it. Read off the glyphs, the row holds exactly 2 . 3 space (8 . 4) beginning at x0 = 268.50, which is where every other three-character value in the column begins, and no minus and no 1 is inked anywhere to its left: the short dash that sits there is the row's axis tick, which the positive rows carry too. The authors' own field_ates_plot.pdf is the same in both respects, so the leading −1 was lost when the figure was drawn rather than when it was composed: the label overflows the plot panel and its −1 is clipped out of the ink, while the text layer keeps it, which is why pdftotext reports −12.3 for a page that shows 2.3 and why a token gate cannot see this defect at all. Nothing about the estimate is in doubt. The article's own introduction prints it as (−12.3 points, SE: 8.4 points) and the deposit gives −12.307692 (8.384118), so the paper disagrees with itself the way entry 2 records for table S2.1. The corrected figure is the maintained pipeline's own, which draws the full value: of the 420 words and numbers it carries, 419 are on the published page and the one exception is the restored −12.3.

Carried, and not corrected

The six entries below, and entry 16 among the reference-list entries, are different in kind from the twelve corrections in this note. Each quotes a passage this edition reprints exactly as the paper published it, and proposes no change to it.

Four of the six are strings the deposited replication archive supplies to the supplement's figures and tables, which print whatever the data hold: a study label misspelled in all eight places the supplement names those two studies, two occupation labels, and a zero whose printed minus sign is floating-point residue rather than a negative estimate. Correcting any of the four on the page would put this edition's floats out of step with the data behind them, and would correct the supplement for a defect that is not the supplement's. Two of the others, entries 18 and 19, are statements in the article's own prose that the paper contradicts elsewhere in its pages: "over 70 papers" against the 51 reports and 89 assessments of the PRISMA diagram, and an earliest study dated 1986 in the introduction and 1983 in the abstract. Neither can be corrected from the deposit, which records neither the search nor the cited 1987 paper, and a replacement number would be the authors' choice and not this edition's. The last, entry 16, is a journal abbreviation carrying a stray letter, set the same way in both of the two places this paper prints its reference list, and it sits with the reference-list entries below.

Each is recorded here because an edition that silently reproduced them would leave a reader unable to tell a faithful reprint from an uncorrected error.

8. The two Ahmed studies are named "Ahmed at al." in all eight places the supplement names them

Figure S4.5: **Ahmed at al.** (14): Sweden

The supplementary materials set "et al." 111 times and "at al." eight, and all eight name one of these two studies: fig. S1.1's two rows, table S2.1's two rows, the captions of figs. S4.5 and S4.6, and fig. S5.62's two rows. It is one string printed eight times rather than eight slips. The deposited data carry the study label in a column of their own, Ahmed at al. (14): Sweden and Ahmed at al. (66): Sweden, and no line of the maintained rewrite spells either of them, so every printing of the misspelling in the supplement is a printing of that one value. The authors' own working figures carry it as well: field_ates_plot.pdf, gender_gradients_plot.pdf and study_manifest.tex in the archived project set it exactly as the published pages do. References 14 and 66, the two studies these labels cite, are set correctly in both printings of the reference list, so nothing here leaves a reader in doubt about which studies are meant.

9. Figure S4.32 misspells an occupation label

industriel production 20.83 (27.88)

The row belongs to Dahl and Krog (84): Denmark, and the label is the occupation as the deposited data spell it: dahl_1_occupation_cates.pdf in the authors' archived project prints the same string. The supplement spells "industrial" correctly where it sets it itself, in "industrial relations officials" and "industrial engineering". This is the same study whose occupation count entry 2 corrects, and the two findings are independent: the count is a number the article contradicts elsewhere in its own pages, and the label is a spelling the data carry.

10. Figure S4.37 misspells an occupation label

indistrial engineers −3.57 (1.55)

The row belongs to Gonzalez et al. (89): Spain, and as with entry 9 the label is the occupation as the deposited data spell it: gonzalez_1_occupation_cates.pdf in the authors' archived project prints the same string. It is the fifteenth occupation row of the panel. Gonzalez et al. is the other of the two studies whose occupation count entry 2 corrects.

11. Figure S4.28 prints a negative zero

shop sales assistants **−0.00** (2.75)

The estimate is zero and the minus sign is arithmetic residue. The maintained rewrite returns −1.57e-14 for this row, and 1.43e-14 for the lower secondary teachers (language) row three rows above it, which the same panel prints as 0.00 with no sign. Both differences in means are exactly zero, and what separates the two printings is which side of zero the floating-point remainder fell on. The sign therefore carries no information about the estimate, and it is the only one of the supplement's five rounded zeros that prints a sign. The authors' own carlsson_1_occupation_cates.pdf sets it the same way, so this is the deposit's arithmetic reaching the page rather than the press's typesetting.

18. The number of papers the PRISMA search is said to have yielded does not match the PRISMA diagram

> The PRISMA search process yielded **over 70 papers** that met our inclusion criteria.

The PRISMA flow diagram, fig. S6.63 (published appendix p. 46), reports 51 reports included and 89 assessed for eligibility. No reading of the diagram gives over 70: neither count reaches it, and the diagram does not say which of its counts the sentence means. The diagram records the screening process, and nothing in the deposited replication archive records that search, so the pipeline has no number to set against the sentence. Entry 5 quotes the same sentence to correct the figure it cites, and reprints "over 70 papers" unchanged on both sides, because a number that cannot be computed is not one that entry can change. No correction is set here either: the replacement would be a choice between 51 and 89, or a third count of the authors' own, made on their behalf. The paper's estimates rest on the studies it reanalyses, which the sentence does not describe, so no conclusion changes.

19. The introduction dates the earliest study to 1986, where the abstract dates the record from 1983

> … who sent 1982 resumes of fictitious men and women applicants to 991 job openings in Victoria, Australia in **1986** and found a -2.4 percentage point effect of being a woman on receiving an invitation to interview (cluster-robust SE: 1.2 points).

The sentence describes Riach and Rich (8), the earliest study in the record. The abstract begins "Since 1983", and the deposit dates that study to 1983 as well: the earliest year on the paper's own evidence map, fig. S2.2, is 1983. The article therefore disagrees with itself by three years about when the first audit was sent. Which date is right is a fact about the cited 1987 paper, not about this deposit or anything the maintained rewrite computes, and this edition does not hold that paper, so there is no token it can set in place of either. The other quantities in the sentence, 1982 resumes and 991 openings, agree with the deposit. The year dates one study's fieldwork and enters no estimate, so no conclusion changes.

The reference list

The two entries below correct the article's reference list. Each was found by an audit that sends every printed entry whole to Crossref and checks the authoritative record back into it, and each is verified against the record named in its note. No citation in the text is affected, and no cited work is missing: what is wrong is how the reference list sets two authors' names.

12. Reference 22 demotes half of an author's surname to an initial

22. M. Carnes, P. G. Devine, **L. B. Manwell**, A. Byars-Winston, E. Fine, …

Corrected:

22. M. Carnes, P. G. Devine, L. Baier Manwell, A. Byars-Winston, E. Fine, …

The third author's surname is the compound Baier Manwell, and the entry reads its first element as a middle initial. Crossref, doi 10.1097/acm.0000000000000552: Carnes, Molly; Devine, Patricia G.; Baier Manwell, Linda; Byars-Winston, Angela. PubMed indexes her the same way, as Baier Manwell L.

13. Reference 65 breaks two surnames beginning "Mc"

65. M. J. Page, **J. E. Mc Kenzie**, P. M. Bossuyt, I. Boutron, …

… E. Mayo-Wilson, **S. M. Donald**, L. A. McGuinness, L. A. Stewart, …

Corrected:

65. M. J. Page, J. E. McKenzie, P. M. Bossuyt, I. Boutron, …

… E. Mayo-Wilson, S. McDonald, L. A. McGuinness, L. A. Stewart, …

Joanne E. McKenzie's surname is set with a space, and Steve McDonald's is split so that its "Mc" reads as a middle initial. Crossref, doi 10.1136/bmj.n71: McKenzie, Joanne E is the second author and McDonald, Steve the nineteenth. The defect is specific to these two names rather than general to the typesetting: the same entry sets L. A. McGuinness correctly three names later, and reference 60 sets M. McPherson correctly.

The reference list, set twice

The supplementary materials reprint the article's reference list verbatim, all 128 entries, so this paper sets the same list twice. Diffed entry by entry, the two printings differ in exactly two places, and they fall on opposite sides: each document is right where the other is wrong. Neither defect is reachable from one document alone. A comparison of an edition against its own published source cannot see them, because a press that sets one list twice can print it two ways and both comparisons agree with their own page. The second printing is the instrument here, as a clip-aware reader was for entry 7.

14. Reference 9 separates two coauthors with a semicolon

> 9. D. **Neumark;** R. J. Bank, K. D. Van Nort, Sex discrimination in restaurant hiring: An audit study. Q. J. Econ. 111, 915–941 (1996).

Corrected:

> 9. D. Neumark, R. J. Bank, K. D. Van Nort, Sex discrimination in restaurant hiring: An audit study. Q. J. Econ. 111, 915–941 (1996).

The entry names three coauthors and separates the first two with a semicolon and the second two with a comma, so it is inconsistent with itself before it is compared with anything. Crossref, doi 10.2307/2946676: David Neumark, Roy J. Bank and Kyle D. Van Nort are three authors of one article, Q. J. Econ. 111, 915-941 (1996). Every one of the other 127 entries in this list separates its authors with commas, and the supplementary materials set this same reference with a comma. The defect is specific to the article's printing of it.

15. Reference 71 in the supplementary materials drops the period after its number

> **71** S. Baert, A.-S. De Pauw, N. Deschacht, Do employer preferences contribute to sticky floors? ILR Rev. 69, 714–736 (2016).

Corrected:

> 71. S. Baert, A.-S. De Pauw, N. Deschacht, Do employer preferences contribute to sticky floors? ILR Rev. 69, 714–736 (2016).

Every other entry in both printings of this list numbers itself with a digit and a period, and the article's own printing of this same reference has one. Read off the line boxes of supplementary p. 53, reference 71 begins at the item edge x0 = 0.088 like its neighbours and its number is followed by a space rather than a period. It is the only one of the 128 that is, which is what makes it a slip rather than a convention.

Entry 16 is carried rather than corrected: both printings set the abbreviation the same way, so the comparison that found entries 14 and 15 finds it agreeing with itself.

16. Reference 123 misabbreviates Social Indicators Research

… a student sample. **Soc. Indic.s Res.** 116, 731–754 (2014).

The journal is Social Indicators Research, abbreviated Soc. Indic. Res., and the entry attaches a stray s to the abbreviation of Indicators. Crossref, doi 10.1007/s11205-013-0329-4: Lieselotte Blommaert, Marcel Coenders and Frank van Tubergen, "Ethnic Discrimination in Recruitment and Decision Makers' Features", Social Indicators Research 116, 731-754 (2014), whose short container title there is Soc Indic Res. Unlike entries 14 and 15, this one is set IDENTICALLY in both printings of the list, so the second printing reaches it and finds the document agreeing with itself: what it is wrong against is the journal's name rather than the volume's other printing. That is what makes it a note here rather than a twelfth correction, since correcting it would put this edition's two reference lists at odds with the two the paper published.

Entry 17 is of a third kind. Both printings set it the same way, so the second printing finds nothing, and what it disagrees with is the page range Crossref records for the article.

17. Reference 83's page range ends one page early, in both printings of the list

Am. J. Sociol. 112, **1297–1338** (2007).

Corrected:

Am. J. Sociol. 112, **1297–1339** (2007).

Crossref, doi 10.1086/511799: Shelley J. Correll, Stephen Benard and In Paik, "Getting a Job: Is There a Motherhood Penalty?", American Journal of Sociology 112, 1297-1339 (2007). The supplementary materials reprint the entry with the same range, so it is corrected in both places.